PROBLEM STATEMENT

In today's digital era, social media platforms such as Instagram, Twitter, Facebook, and YouTube generate a massive volume of user-generated data in the form of posts, comments, likes, shares, and interactions. This data is highly valuable as it reflects user opinions, preferences, and engagement patterns. However, due to its large size, unstructured nature, and continuous generation, analyzing this data using traditional data processing methods becomes inefficient and challenging.

The major problem lies in efficiently processing and analyzing this large-scale social media dataset to extract meaningful insights such as **user sentiment (positive or negative)** and **engagement trends**. Conventional systems are not capable of handling such **high volume, velocity, and variety of data**, which creates a need for scalable and distributed data processing solutions.

To address this problem, this project utilizes the Hadoop ecosystem, which provides a reliable and scalable framework for Big Data processing. The dataset is stored in the Hadoop Distributed File System (HDFS) and processed using the MapReduce programming model. The mapper function is used to classify user comments into **positive and negative sentiments**, while the reducer aggregates the results to generate overall sentiment counts.

Furthermore, Hive is used as a data warehousing tool to perform SQL-like queries on the processed data. This enables easy analysis and helps in deriving insights such as **sentiment distribution and user response trends**.

KEY POINTS

- Large volume of social media data
- Unstructured and complex dataset
- Difficulty in traditional data processing
- Need for scalable Big Data solution
- Use of Hadoop (HDFS + MapReduce + Hive)
- Extraction of sentiment and engagement insights

SUMMARY

This project focuses on solving the problem of analyzing large-scale social media data by using Hadoop technologies to efficiently process, classify, and extract meaningful insights.